Course Syllabus: Back-End Development with .NET

Course Overview: Getting Started

This course equips you with essential back-end development skills using .NET, preparing you to create robust, scalable APIs and applications. By mastering .NET architecture, C# fundamentals, and API integration, you'll gain the expertise to build high-performing systems that meet real-world demands. Leveraging tools like ASP.NET Core and Microsoft Copilot, you'll also streamline complex processes like debugging and middleware integration. This knowledge provides a strong foundation in .NET development, giving you the confidence to tackle diverse back-end projects and succeed in an in-demand field.

Learning Objectives

By the end of this course, you will be able to:

- Describe the features and functionalities of the .NET Framework, the differences between .NET Core and .NET Framework, and common libraries and packages used in .NET development
- Explain the steps to set up the development environment for .NET, create a simple web API with ASP.NET Core, and integrate OpenAPI (Swagger) with ASP.NET Core
- Define the basic syntax and features of C# in the context of .NET, and the importance and process of serialization and deserialization in .NET
- Develop an API project with Microsoft Copilot, including writing, debugging, and implementing API code and middleware components

Module Overview

Module 1: Introduction to .NET and its Architecture

By the end of this module, you will be able to:

- Describe the features and functionalities of the .NET Framework
- Compare .NET Core and .NET Framework
- Define the basic syntax and features of C# in the context of .NET

- Identify common libraries and packages used in .NET development

Graded Quiz: Introduction to .NET and its Architecture

- For this quiz, you will answer questions based on your knowledge of .NET architecture, covering core framework features, .NET setup, C# syntax, and essential libraries.

Module 2: Building Web APIs with ASP.NET Core

By the end of this module, you will be able to:

- Describe the features and benefits of ASP.NET Core for web API development
- Explain the process of creating a simple web API with ASP.NET Core
- Describe routing and attribute routing in ASP.NET Core
- Explain the concept and implementation of dependency injection in ASP.NET Core
- Describe error handling and logging best practices in ASP.NET Core

Graded Quiz: Building Web APIs with ASP.NET Core

- For this quiz, you will answer questions based on your knowledge of ASP.NET Core API development, covering core features, routing, dependency injection, and error-handling best practices.

Module 3: Serialization & Deserialization

By the end of this module, you will be able to:

- Define serialization and its importance in .NET
- Explain how to serialize objects in .NET
- Describe the process of deserializing objects in .NET
- Identify performance considerations for serialization and deserialization
- Explain security best practices for serialization and deserialization

Graded Quiz: Serialization & Deserialization

- For this quiz, you will answer questions based on your knowledge of serialization and deserialization in .NET, including methods, performance considerations, and security best practices.

Module 4: Middleware and OpenAPI

By the end of this module, you will be able to:

- Define middleware and its role in ASP.NET Core applications
- Describe common middleware components used in ASP.NET Core
- Explain how to integrate OpenAPI (Swagger) with ASP.NET Core
- Describe the process of generating API clients using Swagger
- Identify best practices for designing middleware in ASP.NET Core

Graded Quiz: Middleware and OpenAPI

- For this quiz, you will answer questions based on your knowledge of middleware and OpenAPI in ASP.NET Core, including middleware roles, OpenAPI integration, client generation, and design best practices.

Module 5: Using Microsoft Copilot for Developing APIs

By the end of this module, you will be able to:

- Describe the features and functionalities of Microsoft Copilot for API development
- Use Microsoft Copilot to write API code
- Utilize Microsoft Copilot to debug API code
- Implement middleware components using Microsoft Copilot
- Apply Microsoft Copilot in a comprehensive API development project

Project: Using Microsoft Copilot for Developing APIs

- For this project, you will leverage Microsoft Copilot to develop a robust API, applying the skills gained in understanding Copilot's features, writing and debugging API code, and implementing middleware components. You'll design a cohesive API solution incorporating these functionalities, showcasing Copilot's capabilities to streamline development and ensure reliable performance.